Student Performance Analysis Dashboard (Excel)

An interactive Excel dashboard that explores which factors are associated with students' final grades, built with PivotTables, PivotCharts, slicers and formulas.
Dashboard

Question

Which factors (weekly study time, past class failures, school absences, and ambition to pursue higher education) are associated with a student's final grade?

Dataset

- **Source:** UCI Machine Learning Repository, *Student Performance* dataset (maths course file `student-mat.csv`).
- **Size:** 395 students, 33 columns (demographics, family background, study habits, absences, past failures and three grades).
- **Target:** `G3`, the final grade, scored 0 to 20 (a grade of 10 or more is treated as a pass).
- **Citation:** P. Cortez and A. Silva, *Using Data Mining to Predict Secondary School Student Performance*, 2008. Please check the dataset page for the license before reuse.

Tools

Microsoft Excel: Tables, PivotTables, PivotCharts (combo charts), slicers, `IF`, `CHOOSE`, `COUNTIF(S)` and `AVERAGEIFS` formulas.

Method

1. Loaded `student-mat.csv` (semicolon-delimited) into Excel and converted it to a Table (`tblStudents`). Checked for blanks and duplicates (none found).
2. Added helper columns:
   - `Result`: Pass if G3 is 10 or more, otherwise Fail
   - `Band`: Excellent (16+), Good (14 to 15), Satisfactory (10 to 13), At Risk (below 10)
   - `Absence Level`: None (0), Low (1 to 5), High (6+)
   - `Study Time Label`: study-time codes 1 to 4 converted to hour ranges
   - `G3 Recorded`: flags students whose final grade is 0
3. Built PivotTables and PivotCharts for each factor, showing average grade alongside group size.
4. Connected a **G3 Recorded** slicer to all pivots so every chart can be filtered together.
5. Assembled a one-page dashboard with KPI cards, charts and a key-insights summary.

Data quality note

38 students (about 10%) have a final grade of 0, and all of them have zero recorded absences. This most likely means they did not sit the final exam, so they behave like

missing data rather than genuine zero scores. They drag down the "no absences" group in particular (8.44 including them, 12.61 excluding them). All grade averages on the dashboard therefore use the **357 students with a recorded final grade**, and the slicer lets you switch the 38 back in.

Key findings (357 students who sat the final)

Factor	Finding
Weekly study time	Students studying 5+ hours (n = 83) average about 12.6 vs about 11.2 for those studying under 5 hours (n = 274). The over-10-hours group is small (n = 24).
Past class failures	Grades fall as failures rise: 11.94 (0 failures, n = 294), 10.15 (1, n = 40), 8.83 (2, n = 12), 8.27 (3, n = 11).
Absences	Grades fall as absences rise: 12.61 (none, n = 77), 11.65 (1 to 5, n = 134), 10.84 (6+, n = 146).
Higher education	Students who want higher education average 11.60 vs 9.71 for those who do not.
Overall	Average final grade is 11.52 out of 20, and 74.2% of students who sat the final scored 10 or more. About 26% fall in the At Risk band (below 10).

Limitations

- The results show associations, not causes. For example, students who study more may also differ in motivation or prior ability.
- Some groups are small (for example 2 or 3 past failures, or the "no" group for higher education), so those averages are indicative only.
- The data comes from two Portuguese schools and one subject (maths), so the findings may not generalise.

Files

- `Student_Performance_Dashboard.xlsx` : the workbook (`Data` , `Pivots` and `Dashboard` sheets)
- `dashboard.png` : screenshot of the dashboard
- `student-mat.csv` : original dataset, from UCI

How to use

Open the workbook, go to the `Dashboard` sheet, and use the **G3 Recorded** slicer to include or exclude the 38 students with a final grade of 0.